Why the functional-head theory in syntax is probably wrong

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This paper examines the functional-head theory in syntax, i.e. the claim that grammatical meanings are represented as heads in syntax, in much the same way as contentive heads (verbs and nouns). It starts out from the presumption that the functional-head theory can be expected to predict typical head behaviour of grammatical elements, especially with respect to the core head properties: hyperonym status, distributional identity, obligatoriness, idioselection. These general predictions do not seem to be supported by the cross-linguistic evidence, so I go on to discuss additional motivations for the functional-head theory. In practice, the functional-head theory has not been justified in terms of typical head behaviour, but in rather different ways. I present some alternatives to the widely adopted functional-head notation, and then I offer a general discussion of how one might justify general theories in syntax. My conclusion that the functional-head theory appears to be wrong is not very certain, but rather than assume its correctness, future research should formulate it in a clearly testable way and find robust empirical evidence to determine whether it is correct or not.

1. Introduction

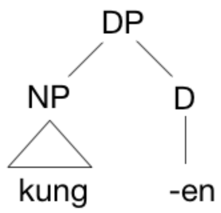
The goal of this paper is to point out that the functional-head theory that has been widely assumed in (mainstream) generative syntax since the 1990s is probably wrong. This is significant because the functional-head theory has often been relied on unquestioningly, and alternatives have not been discussed widely by syntacticians. Discussing and comparing alternative approaches has also been one of the hallmarks of Klaus Welke's work (e.g. Welke 2011; 2019).

By "functional-head theory", I mean the idea in (1) that has been very widely adopted.

(1) Many grammatical ("functional") meanings are represented in the grammar of human languages as heads, in much the same way as contentive heads such as verbs and nouns, as a consequence of innate grammatical knowledge.

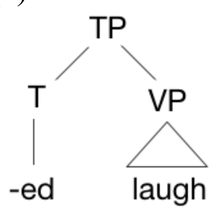
For example, definite markers may be represented as in (2), and tense markers may be represented as in (3). These representations are analogous to representations with the contentive heads V, N and A, as illustrated in (4).

(2)

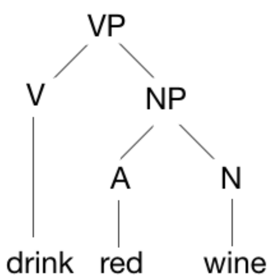


‘the king‘ (Swedish)

(3)



(4)



The functional-head theory has not been given this name before, and it is stated in a somewhat vague way in (1) (“many grammatical meanings”), but anyone who has some familiarity with generative grammar of the 21st century will understand what I mean. The functional-head theory is not normally presented as a coherent theory, but virtually all works in mainstream generative grammar of the last three decades have adopted it as a background assumption, usually without further discussion. For this reason, it is an important question whether the theory is actually true.

2. A representative list of grammatical meanings

Before proceeding further, let us consider a representative list of grammatical meanings which have been claimed to be expressed as functional heads. Within the domain of nominals, some of the most important functional heads are listed in (5).

(5)	D	determiner (article, demonstrative)
	Num	number
	K	case
	Clf	classifier
	Co	coordinator
	Q	quantifier

Within the domain of verbs, some of the most important heads are those in (6).

(6)	v	(little v)
	I(infl)	inflection
	Agr	agreement
	T	tense
	Asp	aspect
	Voice	voice

And within the larger domain of the clause, some of the most important heads are those in (7).

(7)	Pred	predicate
	C	complementizer
	Top	topic
	Foc	focus
	Force	illocutionary force

All of these have often been treated as functional heads since the 1990s, and there are many more that I do not mention here for reasons of space. Even though it is often said by generative linguists that the theory of grammar should be restrictive, there has been no sustained attempt (as far as I know) at limiting the number of functional heads that might be posited. Some authors have even suggested that there may be many dozens of them (e.g. Cinque 1999; Rizzi & Cinque 2016),[1] though this has been very controversial (cf. Chomsky et al. 2019: 251).

3. What the functional-head theory can be expected to predict

There are two ways in which one can think of trees like those in (2)-(4). First, they could simply be notational devices which allow conveniently uniform (binary branching) tree representations but make no further claims about what Human Language is like. It may be that some authors would adopt this stance, but I will not discuss this possible view any further, as I take it for granted that there are many other notational options that are much simpler and much more practical.[2] Moreover, mainstream generative grammar work is often presented with a peculiar rhetoric that emphasizes high explanatory goals (cf. Borsley 2017).

Thus, the interpretation of the trees in (2) and (3) and of the functional-head theory that I will consider here is another one: I will interpret it as claiming that functional heads are part of "universal grammar" (UG), i.e. the innate biocognitive blueprint, "determined by

[1] The situation is thus similar to the situation in optimality theory, where a restrictive view would suggest that there can exist only a limited number of constraints, but in fact hundreds or thousands of constraints have been posited, and OT in phonology has become less and less restrictive over time (van Oostendorp 2022).

[2] For example, prose descriptions as in typical reference grammars (e.g. Haspelmath 1993; McPherson 2012; Berghäll 2015) are much more practical for human users, and dependency treebanks (e.g. Nivre et al. 2018) are much more practical for automatic language processing. Other notations are similarly complex but are known to be more precise and formally rigorous, e.g. HPSG notation (Müller et al. 2021).

genetically encoded properties of our brain, a product of the biological evolution of our cognitive system" (Holmberg 2017: 275). My conclusion will be that there is almost no evidence for the claim that functional heads are part of an innate grammar blueprint, and that what we find in the world's languages is fully compatible with the alternative proposal that the similarities within and across languages are due to similarities of function. (See also Welke (2002) for a functional view of language structure.)

If the contrast between heads and non-heads is part of an innate language blueprint, then we expect heads to be uniform within languages and across languages, at least in a probabilistic way. If they are not generally uniform at least in some way, there does not seem to be any basis for claiming that functional meanings are represented as heads, and the functional-head theory receives no support.

So what are the general properties of heads? How do we recognize a head, and how do we decide whether something is a head if this is not immediately apparent? Given the prominence of the "head" concept in morphosyntax, it is surprisingly difficult to answer these questions. The concept of a "head" has been widely known in generative grammar since the early 1980s (especially since Jackendoff 1977), and following Nichols (1986), the head/dependent contrast has also been used widely by other comparative grammarians (e.g. Dryer 1992; Vincent 1993; Helmbrecht 2001; Lander & Nichols 2020). But despite the widespread adoption of heads in syntax, not much has been said about how heads can be identified and characterized in a general way since the 1980-1990s (when Zwicky 1985, Hudson 1987, Croft 1996 and others discussed the head notion).[3]

So in the absence of recent authoritative characterizations of heads and dependents, I take uncontroversial contentive heads (noun, verb and adjective heads) to have the key properties in (8).[4]

(8)
a. a head is a hyperonym of the larger expression
b. a head has the same syntactic distribution as the larger expression
c. a head is obligatory
d. a head idioselects some of its dependent types
   (i.e. imposes idiosyncratic, nonsemantic selectional requirements)

The semantic hyperonymy relation (a) between heads and the corresponding head-dependent combinations is the most intuitive property that distinguishes heads. It is easy to see that in the pairs in (9), the second element is a kind (or subtype) of the first, i.e. hot weather is a kind of weather, playing in the garden is a kind of playing, and so on. This is a very old criterion (examples (9a-b) are from Jespersen (1924: 96),[5] cited by Zwicky (1985: 2)).

[3] See also Welke (2019: 40; 191) on the head in dependency, constituency and construction grammars.
[4] The properties of heads are not often dealt with in textbooks. I found some discussion in Tallerman (1998: §4.1) and Adger (2003: §3.4), which is quite in line with (8a-d).
[5] Jespersen used the terms *primary* and *secondary*, not *head* and *dependent*. The term *head* seems to have been rare before Jackendoff (1977) (but see Bloomfield 1933: 195), and the term *dependent* was extremely rare before Nichols (1986).

(9)

		superordinate (hyperonym)	subtype (hyponym)
a.	N	*weather*	*hot weather*
b.	A	*hot*	*extremely hot*
c.	V	*playing*	*playing in the garden*
d.	N	*pot*	*flower pot*
e.	A	*blue*	*sky-blue*
f.	N	*roof*	*roof of our house*

Linguists are also generally in agreement that in verb-argument combinations, the verb is the head, rather than the argument, and this can also be understood via the semantic subtype relation. For example, *eating soup* can be said to be a subtype of *eating*, and *looking at the stars* can be said to be a subtype of *looking*. (It works less well with entire clauses, though – is (*they) helped me* a subtype of *helped*, or of *helping*?)

Heads are also said to have the same syntactic distribution as the corresponding larger phrases (8b) (in other words, to be "distributional equivalents"). Thus, *hot weather* occurs in contexts where *weather* could occur, not in contexts where *hot* could occur, *sky-blue* occurs in contexts where *blue* could occur, and so on. This is often expressed by saying that the head determines the category of the larger expression. For example, a noun (N) is the head of a nominal phrase (NP).

Next, heads are often thought to be obligatory, while dependents need not be obligatory (8c). This is clearest with modifiers like *extremely*, in *extremely hot*, or *flower*, in *flower pot*, which are always optional. But arguments are also often optional (e.g. *you eat the soup! eat the soup! eat!*), while the verb is typically obligatory in a clause.

Finally, heads are said to impose idiosyncratic, i.e. nonsemantic, selectional requirements on their dependents.[6] This is called *idioselection* in (8d), and it is intended to subsume Zwicky's (1985) old terms *subcategorization* (the requirement of elements of a certain category) and *government* (the requirement that the subcategorized elements have a certain form, e.g. dative case). Idioselection is best-known from verbs, e.g. *rely*, which idioselects an argument flagged by the preposition *on* (*rely on someone*), or *donate*, which idioselects two object arguments, one flagged by the preposition *to* (*donate something to someone*). But prepositions also idioselect the form of their complement in some Indo-European languages, e.g. Russian *dlja* 'for' (always with a genitive complement) vs. *k* 'toward' (always with a dative complement).

There are a few more head properties that were discussed in the late 20th century literature (Zwicky 1985; Hudson 1987; Bauer 1990; Croft 1996), but I will not discuss them here because none has been widely regarded as a head property outside of this literature.

So for the purposes of the present discussion, I take the four head properties as representative. The implicit claim of the functional-head theory can then be summarized as in (10).

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[6] In the generative tradition, many authors distinguish between *c-selection* (for what was earlier called *subcategorization*, for categorial requirements) and *s-selection* (for semantic requirements, what used to be called *selectional restrictions*) (e.g. Pesetsky 1995: 2). In the European (and HPSG) tradition, the counterpart for verbs is *valency* (e.g. Malchukov & Comrie 2015), which encompasses both "subcategorization" and "government". The term *idioselection* is chosen here because *valency* is best restricted to verbs.

(10) Functional heads such as those in (5)-(7) tend to have the head properties in (8a-d) in the world's languages.

This claim has never been stated explicitly in this way, as far as I know, but the crucial point is that it is necessarily implicit in the practice of using functional heads in generative grammar. If one does not want to hypothesize that (10) is true, one cannot use functional heads as part of the theory, because in that case, there is no reason to say that contentive heads and functional heads are heads in the same sense.[7]

4. Testing the implicit predictions of the functional-head theory

A full-scale test of the functional-head theory cannot be done here, not only because I do not have the resources for it, but also because the scope of the claims in (10) is very vague. However, it is important to spell out how one could test the theory if the claims were made more precise, because it must be testable if it to be taken seriously as a scientific theory. The literature usually presents the theory without much justification (in textbooks and handbook articles), but we do not know whether it is true. Maybe others will take up the challenge and carry out a full or partial test after making the claims more precise.

4.1. Testing hyperonymy

In order to test the claim that functional heads such as articles, number markers, tense markers, voice markers, complementizers and illocutionary force markers are hyperonyms of the phrases that they are said to head, one needs to compare the meanings of the functional elements to those of the corresponding head-dependent combinations. Are the expressions in the right-hand column in (11) hyponyms, or subtypes, of the functional heads?

(11)

		superordinate (hyperonym)	subtype (hyponym)
a.	D	*the*	*the weather*
b.	Num	*-s*	*garden-s*
c.	T	*-ed*	*play-ed* (past tense form)
d.	Voice	*-en*	*tak-en* (passive voice form)
e.	C	*that*	*that we are playing*
f.	Force	*czy*	*czy widzisz?* 'do you see?' (Polish; *czy* is a question particle)

Such a comparison may seem difficult, but we need to describe the meanings of all forms anyway, and there are many specialists of compositional semantics who should be able to carry out such comparisons.

[7] It cannot be excluded, of course, that there is some other property, unrelated to (8a-d), that is shared by contentive heads and functional heads. However, the literature does not contain any extensive discussion of other properties.

For some of the functional elements, it seems clear at first blush that there cannot be a hyperonymy relationship: *Garden-s* is a kind of thing, not a kind of plurality, and *play-ed* is a kind of action, not a kind of temporality. And the passive form *tak-en* denotes a kind of action as well (presented from a different perspective than in an active construction), not a kind of "Voice". The same would seem to go for most of the other elements that we saw in (5)-(7) (case, classifier, agreement, aspect, topic, focus).[8]

However, it could be that a more sophisticated semantic analysis will show a different picture. And in fact, Hudson (1987) argues that determiners, auxiliaries and complementizers are "semantic functors", and that this is the correct way of expressing the original hyperonymy idea. He suggests that a phrase with an auxiliary such as *may control those penguins* refers not to a kind of action, but to a kind of possibility. This is not too different from saying that *play-ed* is a kind of temporality, though Hudson does not consider affixal functional elements.

Hudson's suggestions on the semantics have not been followed up on, and the suggestions in the last paragraph are extremely tentative, so for now, I conclude that functional elements are probably not hyperonyms of their phrases.

Quite apart from hyperonymy considerations, an important development is that many linguists have argued for functional heads in syntax on the basis of semantic considerations (see, e.g., Larson 2019: §1.3 for determiners). In their comparison of HPSG and the Chomskyan approach, Borsley & Müller (2021: §4.16) discuss this under the rubric of "syntactification of semantic categories". However, these considerations are not usually connected to any specific properties of contentive heads, so they are not relevant to testing the functional-head theory. (Recall from (1) above that this theory claims that functional heads are heads in the same way as contentive heads.)

4.2. Testing distributional equivalence

Are functional heads the distributional equivalents of the corresponding phrases? In general, this is not the case. Functional heads are grammatical markers, and as such they usually do not occur on their own at all. Number markers like *-s* in *garden-s*, tense markers like *-ed* in *play-ed*, and complementizers like *that* (in *that we are playing*) are bound forms that cannot occur independently of their hosts. Some languages allow auxiliaries to occur without a verb host (e.g. English, but not French, Croft 1996: 43), and in many languages, demonstratives can occur on their own. But by and large, functional elements do not have the same distributions as the larger expressions of which they are part.

This was discussed by Zwicky (1985: §2.5.1), with a similar conclusion, but Hudson (1987: §5-6) argued that if one changes the assumptions slightly, one can say at least for complementizers and articles/determiners that they have the same distribution as clauses and nominals, respectively: If one includes adverbial subordinators like *after* and *since* in the class of complementizers, then some of them do indeed occur on their own as adverbs (cf. *I haven't heard from her since she moved away* vs. *I haven't heard from her since*).

[8] Early in my career, I had argued that affixes are often head-like because they arise from heads diachronically (Haspelmath 1992). However, I did not claim that this applies to all affixes, and I had no clear understanding of "head" at the time. (See also Welke (1995), who argued against the head status of derivational affixes from a different perspective.)

And if one says that articles really belong to the same class as personal pronouns, then some members of them do occur on their own.

But while Hudson may have convinced himself that this is sufficient to say that articles and complementizers are distributional equivalents of nominals and clauses, respectively, it cannot serve as a general test of the hypothesis. To test the hypothesis in an objective and rigorous way, we need to make clear precisely what forms are subsumed under the classes D, Q, T, Asp, Top, C and so on, and one needs to examine for all of them whether they occur in the same environments as the corresponding phrases. As long as this has not been shown, at least for a significant majority of the cases, the hypothesis cannot be taken to be confirmed, and it does not look promising.

4.3. Testing obligatoriness

Are functional heads generally obligatory, while their dependents are optional? Again, in general this is not the case. I will give just a few examples. In most languages, clauses ("CPs") can occur without complementizers, many "TPs" can occur without a tense marker (cf. *we play-ed*, *we play-Ø*), nominals in most languages ("DPs") can occur without a determiner, number markers for singular are found in few languages, and many languages lack markers of illocutionary force entirely. On the contrary, it is generally the sister constituents of the functional elements that are obligatory (as noted by Abney 1987; Muysken 2008: 62).

What exactly is meant by obligatoriness may not be quite clear, so a rigorous test of this hypothesis would have to sharpen this notion (for example, does contextual ellipsis count? if so, then even verbs are not obligatory in VPs, because they can be omitted in gapping). I noted earlier that non-obligatoriness is particularly characteristic of modifiers (as in *extremely hot*, or *roof of our house*), while argument dependents are often not optional at all (cf. *put it here!* vs. **put!*). And it may be important that inflectional exponents are often thought to be obligatory by definition, so that, for example, the fact that a singular-plural contrast is an inflectional paradigm means that there must be a "singular exponent" (even if it is zero).[9]

Perhaps obligatoriness is not a good head criterion (and Zwicky 1985: 13 noted that it is closely related to the distributional equivalence criterion). But we have few good head criteria, and without such criteria, the functional-head theory cannot be tested, so I wanted at least to mention this criterion here.

4.4. Testing idioselection

Does a functional element idioselect its sister constituent? In other words, does it impose idiosyncratic, nonsemantic selectional requirements on it, in an asymmetric way? To answer this question, we first need to make the notion of idioselection more precise. I define this new term provisionally as in (12).

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[9] However, this criterion for inflection seems to be too narrowly based on Indo-European languages, where number and tense markers are typically obligatory. Elsewhere in the world, such markers are very often optional (as also noted in Haspelmath 2024: §5.7).

(12) A member of a form-class A idioselects a form-class B if the two are sister constituents and if the choice of elements forms from A forces the choice of different elements from B.[10]

This definition is intended to be a generalization of (part of) the notion of *valency*, which is generally reserved for verbs, and which was illustrated earlier. The fact that verbs idioselect their arguments (= impose valency requirements on them) is well-known, but the phenomenon of idioselection is broader. In some Indo-European languages, prepositions also idioselect their complements. Nouns are not often taken to idioselect their dependents, but in fact, many languages around the world have different constructions for adnominal possessors, depending on the possessed noun (e.g. Maltese *ktieb ta' Luisa* 'book of Luisa' vs. body-part terms: *id Luisa* 'Luisa's hand'). This can be regarded as a kind of "valency" of nouns. It is also clear that when it comes to the combination with complement clauses, different verbs require different forms of clauses. Thus, *stop* idioselects an *-ing* complement (*she stopped working),* while *cease* selects a *to*-infinitive complement (*she ceased to work*). Valency is a property of the head verb, and the idioselection properties of prepositions and nouns may likewise be taken to derive from their head status.[11]

So do function elements generally idioselect their co-constituents? To some extent, the answer is yes: It is widely accepted that in English (and other European languages), complementizers and auxiliaries idioselect their sister constituents in the sense that they require a particular form of the verb in English (e.g. *for her to go* vs. *that she goes*; *will go* vs. *has gone* vs. *is going).* But determiners such as articles or demonstratives do not require a particular form of the noun, in English or other languages. So it might be that determiners are not functional heads, whereas complementizers and auxiliaries are (in English) (cf. Bruening et al. 2018).

But what can we say about functional elements (like those given in (5)-(7) above) in general? Are English complementizers and auxiliaries typical of functional elements in general, and are only determiners atypical? This question is very difficult to answer, because we do not have large-scale surveys of complementizers and their behaviour in the world's languages. And "auxiliary" is not even a well-defined concept that can be readily applied outside of Indo-European languages. But once we broaden our horizon to other functional elements and other languages, we see that it is not difficult to find cases where it is the contentive element that idioselects the functional element, rather than the other way round, e.g.

(13) a. auxiliary selection:
French verbs select the type of auxiliary they combine with (*être* vs. *avoir*, e.g. *il est allé* 'he has gone' vs. *il a vu* he has seen')

b. tense marker selection:

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[10] I use the term "form-class" (Bloomfield 1933) here because it is more precise than the term "(morpho-)syntactic category. A form-class is a class of forms that behave alike in some construction.

[11] But note that while verbs, nouns and (to a limited extent) adjectives may idioselect the forms of their arguments, they do not determine the forms of their modifiers. The forms of adverbial expressions, the forms of adjectives, and the forms of degree adverbs do not normally depend on particular verbs, nouns and adjectives.

Latin verbs select the type of future-tense marker they combine with (*-bi* vs. *-e*, e.g. *lauda-bi-t* 'she will praise', *audi-e-t* 'she will hear')

c. complementizer selection:
Irish verb forms select the type of 'that' complementizer of their clause: present-tense verbs take *go*, while past-tense verbs take *gur*

d. number marker selection:
German nouns select the type of plural marker they combine with (*-e* vs. *-er* vs. *-en*, e.g. *Tag-e* 'days', *Büch-er* 'books', *Staat-en* 'states')

e. case marker selection:
Russian nouns select the type of case suffix they combine with (e.g. Dative *-u* vs. *-e*, e.g. *volk-u* 'to the wolf', *sobak-e* 'to the dog')

f. quantifier selection:
English nouns select the type of multal quantifier they combine with (*many* vs. *much*, e.g. *many euros*, *much money*)

g. classifier selection:[12]
Mandarin Chinese nouns select the type of numeral classifier they combine with, e.g. *yī běn shū* 'one book', *yī zhī māo* 'one cat'

h. article selection
Tagalog nouns select the type of article they combine with (*ang* for common nouns, *si* for proper names, e.g. *ang isda* 'the fish' vs. *si Pedro*

i. predicate marker (= copula) selection
Jamaican predicates select the type of copula they combine with (*a* for nominal predicates, *de* for locative predicates, e.g. *mi a bwai* 'I am a boy', *de de ya* 'they are here')[13]

j. coordinator selection
Dagbani phrases select the coordinator they combine with (*mini* for coordinating nominals, *ka* for cordinating verbs; Olawsky 1999: 44, 51; see Haspelmath 2005)

I conclude in a preliminary way that there is no evidence that it is a general property of functional elements that they idioselect the form-class of their sister constituents, rather

[12] Classifiers are typically treated together with gender markers under the heading of "nominal classification", so that one might treat noun-classifier dependencies as "agreement" (though this is not often done). I do not consider stereotypical agreement cases here. Bruening et al. (2018: 7) do treat gender agreement of adnominal modifiers in Spanish as a subtype of "selection", but we cannot say that all cases of "agreement" are idioselection (e.g. verbs agreeing in number with their objects) in the sense of (12), because then there would be clear cases of non-heads determining the form of heads (see (14) below).
[13] http://www.jumieka.com/grama.html

than the other way round. There does not seem to be any basis for generalizing from English auxiliaries and complementizers to all functional elements.

And it is not even clear that idioselection as defined in (12) is a very robust property of verbal and nominal contentive heads. In some languages, the form of the verb is different depending on what kind of object it combines with (full nominal vs. pronominal, e.g. Hausa), and the form of the preposition may be dependent on the type of complement it combines with (full nominal vs. pronominal, e.g. Coptic).[14] Moreover, as was hinted in note 8, the notion of idioselection is not a very natural one, because it must actually be defined as in (14), to exclude "agreement".

(14) A member of a form-class A idioselects a form-class B if the two are sister constituents and if the choice of different forms from A forces the choice of different forms from B – unless B "agrees" with A.

If "agreement" were included as a form of idioselection, then it would no longer be intuitively correct to say that contentive heads idioselect their dependents, rather than the other way round, because verbs often "agree" with their argument nominals.

There is one generalization, however, that we can make quite confidently:

(15) Idioselection always involves functional elements, either in the sense that contentive elements idioselect different types of functional elements, or in the sense that the idioselected forms differ in their marking properties (= in the functional elements that they contain).

So what we do not find is idioselection of a nonsemantic subclass of nouns by some class of verbs, or the idioselection of a nonsemantic subclass of adjectives by some class of nouns. One could imagine a language that has two subclasses of nouns (class C and class D), where some verbs only take class C nouns as objects, while other verbs only take class D nouns as objects. This is not what languages do – idiosyncratic selection is strongly associated with functional elements. Thus, verbal valency determines the case-marking of nominals, not the subtype of noun, and complement-taking verbs (e.g. *stop working* vs. *cease to work*) determine the nominalizer (*-ing* vs. *to*) of the complement clause, not the subtype of verb.

But this does not mean that case-markers and nominalizers must be functional heads, because the head property in (8d) says that the idioselector is a head, not that the idioselected element is a head.[15]

4.5. Conclusion of this section

I conclude that there is very little evidence that functional elements such as those in (5)-(7) generally behave like contentive heads with respect to the four head criteria of (8a-d). This

[14] For Hausa, see Newman (2000), and for Coptic, see Layton (2004).

[15] Some authors say or presuppose that the markers of idioselected forms must be heads (e.g. Bruening et al. 2018). But this has no basis in the properties of contentive heads, so it is unrelated to the functional-head theory as formulated in (1). It amounts to a very different (and unclear) notion of "head" that I do not consider here.

tentative finding is a strong reason to suspect that the functional-head theory is probably wrong. (It has been suggested to me that I should say "unwarranted" instead of "probably wrong"; but if a particular claim is not very likely to begin with, I take the absence of evidence for it as indicating that it is probably wrong. As stated in (1), the theory makes reference to innate grammatical knowledge, which is inherently unlikely.)

Of course, the fact that I have not found such evidence does not mean that it cannot be found, but the main purpose of this exercise was to point out that further research is needed if we want to make claims of the sort that have often been made. There is at the moment no basis for assuming that the functional-head theory is probably right, and for using it in teaching syntax to students as if it were somehow established knowledge (as seems to be done by many university instructors throughout the world).

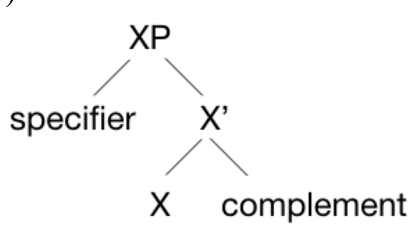
Of course, there is no reason not to discuss this theory with advanced students, but there is no reason, for example, to use this highly tentative theory for studies of language processing or language acquisition. We simply do not know enough about syntax at the present stage in order to trust the generalization that is at the heart of the functional-head theory.

5. What motivated the functional-head theory

In the preceding three sections, I have focused on possible empirical tests of the functional-head theory, and I have come to a preliminary negative conclusion, which may surprise some readers. But experienced readers will not be surprised at all, because the functional-head theory was never justified and motivated in the terms in which I approached it so far. So at this point, I will consider the motivations given by proponents of the theory.

I suggest here that the main reason for claiming that functional elements are generally heads is the widespread idea that all phrases should conform to the X-bar schema in (16) (a version of which was first proposed by Jackendoff (1977) for English syntax).

(16)



Many authors in the Chomskyan tradition have assumed that both the specifier and the complement have to be full phrases ("maximal projections") and cannot be simple words or grammatical markers.[16] That this was part of the motivation for positing functional heads is clear in Abney's (1987) famous argument that nominal expressions are DPs, not NPs. If a prenominal determiner is a modifier rather than a head in English, it does not fit into this

[16] See also Welke (2011: 40) for a similar point of criticism. However, as Adam Tallman reminds me (p.c.), what I say in the text probably describes the situation of the 1990s, during which the functional-head theory became prevalent. It is no longer widely assumed, so that the current situation is still more complex and harder to understand.

schema. According to Abney, "D is no longer defective with respect to X-bar theory, but projects a phrasal node, and takes a complement, like other categories" (1987: 75). This can be seen in example (2) above from Swedish, where the determiner *-en* projects the phrasal node DP and takes *kung-* as a complement.

Until the 1980s, it had been common in generative syntax to distinguish between "major categories" and "minor categories", and it was accepted that the two types of elements behaved differently. Complementizers and auxiliaries in English were not taken as heads of their phrases (this can still be seen in textbooks like Akmajian et al. (2010), which reflect the situation around 1980). But the situation changed abruptly in 1986, when Chomsky proposed that a clause with a complementizer should be called a CP, and the smaller constituent with (finite)verbal inflection should be called IP (a projection of Infl/I, later renamed T).[17]

How does Chomsky justify this change? Here is a quotation from the 1986 book which is at the origin of the idea of functional heads.

> Chomsky (1986: 2)
> "Does this system extend to the nonlexical categories as well? Evidently, the optimal hypothesis is that it does. Let us assume this to be correct. Then the clausal categories conventionally labeled S and S' might be IP and CP, respectively, where I = INFL and C = complementizer, as in (2):
>
> (2) a. S = IP = [NP [I' [VP V . . .]]]
> b. S'= CP = [... [C' C IP]]"

This formulation is highly tentative, and is justified merely by saying that this would be "the optimal hypothesis", which is then assumed to be correct. So there is no attempt to provide empirical support for this hypothesis,[18] and it appears that what I did in (3)-(4) would not have occurred to Chomsky. (I will discuss why this might be the case in the next section.)

In the subsequent literature and in the textbooks, we find a similar pattern. Instead of motivating functional heads on their own merits, the idea of functional heads is said to follow from the attractiveness of having a uniform X-bar schema, e.g.

> Leffel & Bouchard (1991: 5)
> "With the suggestion that functional elements such as Infl and Comp were heads came the apparently natural extension of head status to other functional entities such as determiners and auxiliaries and degree expressions... Abney's (1987) theory applied X-bar theory in the same general fashion to all functional heads."

[17] Abney (1987: 75) makes explicit reference to Chomsky (1986): "This is in keeping with the analysis of I and C which has emerged in recent years (Chomsky 1986), in which I and C are taken to participate fully in the X-bar system. In fact, the Det-as-head analysis is almost forced if we wish to suppose generally that "non-lexical" categories are not defective with respect to X-bar theory."

[18] The lack of empirical evidence for this new uniformity was noted by some prominent commentators in the 1990s, in particular Lightfoot (1990) and Anderson (1999). Anderson wrote that Chomsky's argumentation struck him "as a rather stunning logical leap into the abyss, but it quickly led to a proliferation of structural articulation and a reformulation of the nature of syntactic structure for which the empirical foundation always seemed rather weak..." (1999: 113).

Another version of the same line of thinking is the following:

> McCloskey (1997: 200):
> "The development of the IP model of sentence-structure in the 1980s ... provided a model of clausal structure of sufficient abstraction and flexibility that it could lay some plausible claim to cross-linguistic generality. The analysis involved two crucial elements. The first is the idea that the sentence is universally a projection of inflectional information (tense, person, number and so on) and the second is that it is a regular X-bar projection. This conception was flexible enough to handle clauses built around verbal and non-verbal predicates (...), and it brought with it the restricted flexibility of X-bar theory to account for the known range of attested clause types."

We find the same kind of justification in all the 21st century textbooks on generative syntax, either in terms of X-bar theory, or in more recent parlance, in terms of "constraints on Merge" (e.g. Radford 2004: 72; Carnie 2008: 128;; Freidin 2012: 93; Sportiche et al. 2014: §6.4), if they give any justification art all.[19]

It is sometimes recognized explicitly that this kind of justification is not empirical, but "conceptual". For example, Koeneman & Zeijlstra (2017: 102) say:

> "If *the* is a functional element, we can push for the same analysis that we ended up with for clauses, namely one in which a lexical domain is embedded under a functional domain. By doing this, we provide a conceptual argument for a DP analysis: it allows an elegant parallel with the structure of sentences."

To be sure, the idea that our innate grammar blueprint has only a small number of components is attractive and maybe also motivated by evolutionary considerations (because of Darwin's Problem, we do not want to posit too many innate building blocks; Berwick & Chomsky 2016). But as Culicover & Jackendoff (2005: 4) have noted, Occam's Razor must be applied to entire systems, not merely to underlying structures. And as everyone with some experience in mainstream generative syntax knows, the actual derivations are often of forbidding complexity (e.g. Borsley & Müller 2023: §4.1). Thus, the motivation for the functional-head theory as it was developed in the 1980s and 1990s does not lie in simplicity or elegance in any absolute sense, but in a persistent attempt to maintain the simple X-bar schema (which later morphed into "Merge").

Finally, I should briefly mention the role of functional heads for other purposes, in particular for linearization. In Pollock's (1989) influential paper on the positional properties of French verbs and negation markers, the role of functional heads such as Agr and Neg seems to be primarily to regulate word order by allowing various movement options that were otherwise prohibited. This sort of syntactic analysis became very common in the 1990s, and it is now commonplace to posit abstract functional categories on the basis of phenomena that have little to do with functional elements. Adger (2003: 165) is quite clear about this role of functional heads:

[19] Van Gelderen (2017: 39) does not justify the head status of functional elements and merely tells her readers that "it has been shown how lexical categories, such as N, project to a phrase. In the later 1980s, grammatical categories, such as D(eterminer), C(omplementizer), and T(ense) also came to be seen as the heads of phrases. So, a C such as that is the head of a CP..."

> "Now functional categories serve not only to add non-thematic semantics on to lexical categories, but also to provide positions in the structure of clauses and phrases which relate to lexical categories in various ways."

But why do such positions need to be "provided" in the first place – why aren't they simply there? And why does one need new (and purely abstract) functional heads such as "little *v*", for purposes of case or role assignment (as if these were not perfectly ordinary phenomena of grammar that do not present any puzzle)? It seems that the idea of the X-bar schema in (16) as the core part of the innate grammar blueprint has been extremely powerful, and all other considerations were subordinated to it (see also Müller 2016: §4.6). I have not seen any justification for the fundamental role of this schema in the literature, but there are other strong uniformity assumptions that have been influential in generative grammar over the last few decades (as discussed extensively by Culicover & Jackendoff 2005: Ch. 2-3). Again and again, linguists have adopted seemingly simplifying assumptions on the basis of uniformity considerations, and have then built highly complex analyses on top of these, thus largely nullifying the initial simplicity gains.[20]

Is this a promising way to construct a general theory in syntax? In §7 I offer some general methodological considerations about theory-building, but first we ask what kind of notation one could use instead of the functional-head notation.

6. How the functional-head notation could be replaced

Some readers will ask what I propose instead of the functional-head theory. The answer is simple: agnosticism. If we have no good reasons to believe that a particular hypothesis or theory is correct, we should leave it aside and pursue other goals.[21]

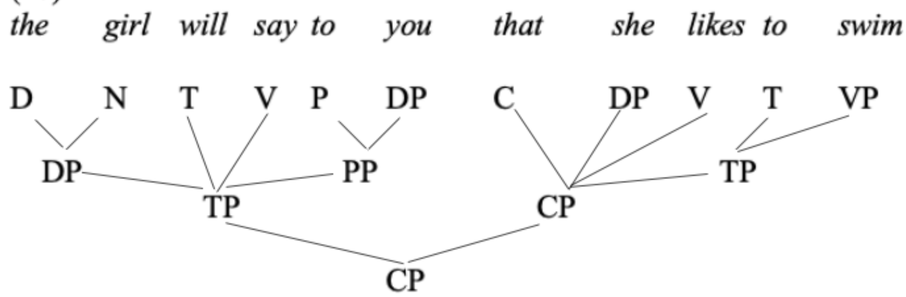
But of course, we need a notation for syntax, and as I mentioned earlier, many linguists do not seem to be committed to the functional-head theory but use the functional-head notation merely as a matter of convenience, because it is widely practiced by others (an example is Anderson & Lightfoot (2002); recall from note 18 that both authors had expressed very strong reservations about the CP-IP system). But using a notation that is strongly associated with a discredited theory is confusing, especially in textbooks and other work in non-specialist contexts. Many linguists do not distinguish properly between theory and notation, so if one does not want to commit oneself to a contentious theory, one should use neutral notation.

A simple way of avoiding the contentious functional-head notation would be to simply place the contentive elements in the highest position of their phrase. Then one would have something like (18) instead of something like (17) (both are simplified, because many of the details do not matter).

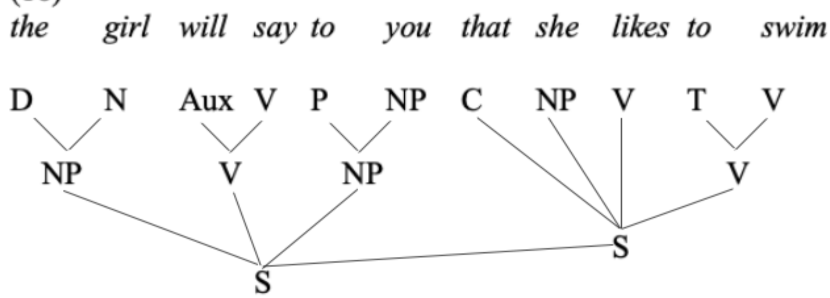
[20] An extreme conclusion that one could draw from the widespread occurrence of "null heads" would be that all functional heads are null, and this was in fact proposed (in all seriousness) by Kayne (2016).

[21] One well-known way of proceeding if a traditional notion does not seem to give the results that one first expected from it is to decompose it into more basic notions (see, e.g., McCloskey 1997 on decomposing subjects, and Bickel 2007 more generally on fine-grained variables). Curiously, it has rarely been proposed that the traditional "head" concept must be decomposed into more basic notions (but see Zwicky 1993 for a very interesting proposal).

(17)



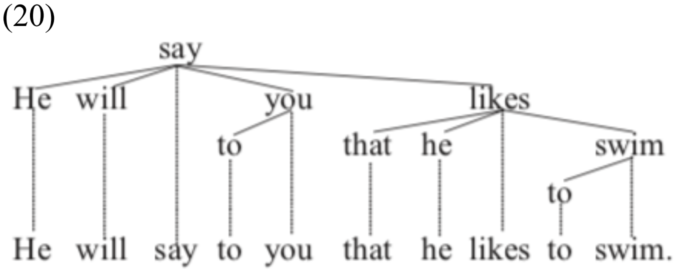
(18)



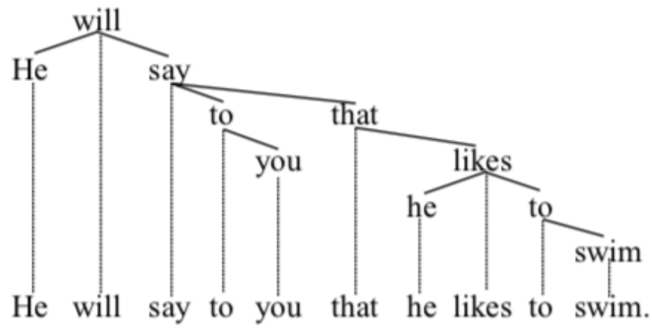
The diagram in (18) does not look very different from pre-1986 generative syntax. Maybe the notation looks old-fashioned, but it does not purport to make claims that are unlikely to be true, so it is a much better notation (it does not even make the questionable claim that English finite clauses have a VP, cf. Osborne & Groß 2016). A recent textbook that does use a notation of this sort is Dawson & Phelan (2016: §5.3). Some readers may wonder about the ternary branching, which went out of fashion in generative syntax in the 1980s. However, there are no empirical reasons to think that syntax generally lacks ternary structures, and the rejection of ternary branching was based on a philosophical uniformity assumption (see Culicover & Jackendoff 2005: Ch. 4).

Another possibility is to use dependency trees, instead of constituency trees. Dependency grammar is getting more and more important in applied domains such as computational linguistics (cf. Müller 2016: Chapter 11; de Marneffe & Nivre 2019), so it may be best to switch to dependency representations if one is not committed to a particular vision of the innate building blocks (see now also Gibson 2025). A slight problem, however, is the fact that there are two ways of representing function words in dependency grammar (see also Welke 2011: 31). The two options shown in (20) and (21) were recently discussed by Osborne & Gerdes (2019).

(20)



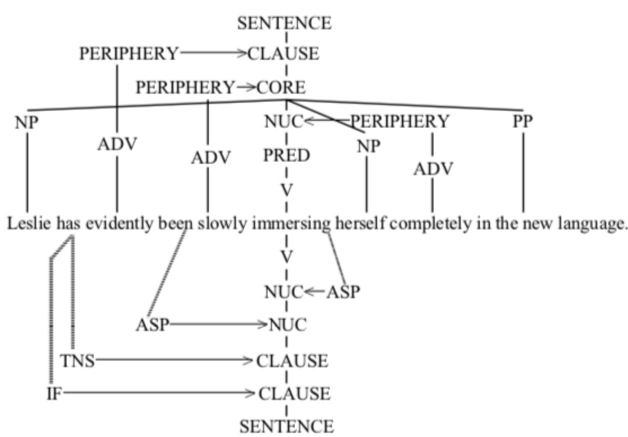
(21)



But recall that the present section is about notation, not about theory. Notation is needed in syntax primarily to visualize hierarchical structures that are not immediately apparent, especially in teaching contexts. The choice between (20) and (21) will not matter much in many contexts.

Finally, one could use some kind of "multidimensional" (Carnie 2008: Ch. 10) notation. For example, Van Valin's RRG (Role and Reference Grammar) uses a notation in terms of a special "operator projection" (shown in a distinct tree-like notation below the example sentence), as illustrated by the example in (21) (Van Valin 2005: 22).

(21)



Such a notation may look complicated at first blush, but note that it is the only representation of the sentence, i.e. there are no additional derivational steps. So the overall complexity is probably much smaller than an equally comprehensive generative grammar representation that involves multiple movement operations.

These examples should suffice as illustrations of the kinds of alternative notations that linguists could adopt if they did not want to be seen to endorse a contentious theory for which there is very little evidence (and which is thus likely to be wrong).

7. What kind of justification is needed for a general theory in syntax

Some readers who are not convinced by my earlier arguments will wonder whether the functional-head theory could not be true after all, but justified by further considerations that I have not mentioned yet. This is a valid objection, and there is a very large literature that has made many points about functional heads that I was unable to touch upon here. So before proceeding, we need to be very clear about kinds of justifications of general theories in linguistics (particularly in syntax).

The term "theory" is used in a variety of ways, so it needs to be clear that I am not talking about theories in the sense of language-particular grammars,[22] but only about general theories that provide explanations that are relevant for all languages (or for Human Language in general). Most works on functional heads consider only a single language (or at most a few languages), so how can they contribute to a general theory?

There are two ways of building a general theory of syntax, what I have called the "non-aprioristic" approach and the "restrictivist" approach (Haspelmath 2014). In the non-aprioristic approach, theories are based on empirical findings from a wide variety of languages, which may be explained in a variety of ways (as, for example, in Croft 2003, Dixon 2010-2012 and Hawkins 2014). In the restrictivist approach, which is adopted by generative grammar, theories are based on the idea that there must be a restrictive set of innate building blocks (formal and substantive universals, Chomsky 1965) that allow language acquisition in the face of the poverty of the stimulus (e.g. Lasnik & Lidz 2016).

Fortunately, despite all the disagreements and lack of mutual comprehension, there does seem to be some kind of consensus about this: Both the non-aprioristic ("Greenbergian") and the restrictivist ("Chomskyan") approach are trying to understand limits on cross-linguistic diversity, but in different ways.[23] The two approaches need not be mutually incompatible, as some phenomena may be explainable in a functional perspective, while others are explained by innate building blocks.

There are two ways in which research on a single language, as is typical of generative syntax, can contribute to a general theory of syntax:[24] On the one hand, they could be justified by noting that language acquisition is impossible without a particular element of the theory. In the case of the functional-head theory, this can be ruled out: Nobody has proposed that the existence of functional heads (as opposed to, e.g., functional elements in "specifier" positions) facilitates language acquisition, let alone is required by it.

On the other hand, restrictivist theories can be justified to the extent that they provide better and better sets of universal innate building blocks for the analysis of all languages,

[22] For example, Adger (2003: 11) says that "a grammar is a linguist's explicit theory of a speaker's tacit knowledge of their language". Such language-particular (or speaker-particular) theories are not relevant here, because I am only considering the functional-head theory in its application to languages in general (see (1)).

[23] Cf. Holmberg (2017: 275-276), who describes the two approaches in very similar terms: "Behind all the variation exhibited by the languages of the world, there would be formal properties that are common to all languages because they are determined by genetically encoded properties of our brain... In [Greenbergian] typology, though, there is no commitment to the idea that the universals are ultimately genetically encoded... They are generalizations; their explanation may vary from case to case. "

[24] These two possibilities are also given by Anderson (2008: 795) in his programmatic paper on the logical structure of linguistic theory: "There are two primary ways in which we infer the nature of Language from the properties of individual languages: arguments from the poverty of the stimulus, and the search for universals that characterize every natural language."

allowing linguists both to describe languages elegantly and to explain limits on cross-linguistic variation, as noted, for example here:

> "The parametric model is a very powerful model of both linguistic diversity and language universals... It makes predictions about language typology: parameters make predictions about (possible) language types ... Furthermore, it draws research on different languages together as part of a general enterprise of discovering the precise nature of UG: we can discover properties of the English grammatical system (a particular set of parameter values) by investigating Chinese (or any other language), without knowing a word of English at all (and vice versa, of course)." (Huang & Roberts 2016)

This approach (known by the name "principles and parameters") is similar to 19th century chemistry, where scientists ended up with a set of pre-established building blocks of all stuff (the Mendeleyevian Periodic Table of Elements; Baker 2001). The study of a particular language (such as English) is analogous to the study of a particular chemical compound, and we hope that by studying many different phenomena in different languages, we will eventually arrive at the innate building blocks (e.g. Baker 2010; Huang & Roberts 2016; Holmberg 2017).

But how do we know whether a particular proposal of innate building blocks is right or not? How can we tell whether we are making progress?

There does not seem to be a general answer to this – it appears that we simply have to try out various possibilities and see which give the best results. But crucially, esthetic or "conceptual" considerations (like those that motivated the functional-head proposals initially) cannot play a role in deciding on competing proposals. The best results are those that bring us closest to the goals of explaining acquisition, limits on cross-linguistic diversity, and allowing good language-particular analyses. But this is extremely hard to measure, as the space of possible hypotheses is enormous. As Kiparsky (2018: 54) put it recently: "reconciling typologies with descriptive frameworks and the analyses dictated by them can involve a labyrinth of choices".

And even though the functional-head theory has reduced the kinds of combinatory possibilities in a certain way, there is no progress as long as there is an unlimited number of possible functional heads, allowing a very large number of impossible languages to be described along with all the possible languages. McCloskey (1997, cited earlier) talks about "restricted flexibility", but as long as there is no clear sense of what kinds of logically possible languages are ruled out by the theory, it is not easy to see any progress.

There was some hope of identifying cross-linguistic universals that can be explained by the principles and parameters approach in the 1990s, as illustrated by Ouhalla (1991), Roberts (1996) and Baker (2001), but those hopes seem to have been given up largely in the 21st century. Bošković (2008) was perhaps the last prominent attempt to link functional heads to actual cross-linguistic generalizations (by arguing that there is an "NP vs. DP parameter", which is associated with many other properties), but his proposals were not widely accepted in the generative literature.

The absence of a clear methodology for making progress may explain why generative linguists often resort to "conceptual" arguments and considerations of "elegance". But elegance is not a substitute for truth, and in any event, there are many different ways in which a theory can be elegant. In the absence of clear consequences for predictions of possible language types, I conclude that the functional-head theory receives no support from the restrictivist approach.

8. Conclusion

The general tentative conclusion of this paper is that the functional-head theory in syntax is probably wrong. This conclusion is not very certain, because the empirical considerations of §3-4 were very preliminary, and the general methodological considerations of the §7 merely found that the theory has so far received no clear support from the results of stimulus poverty or principles and parameters research. Of course, by no means did I demonstrate that the theory is wrong – merely that there is very little evidence for it, if any.

However, if a scientific theory is very much in doubt, it is a safe strategy not to assume that it is true. But unfortunately, linguists often take the functional-head theory for granted: The theory is presented in entry-level textbooks (e.g. Fromkin & Rodman 2017: Ch. 3), thus reinforcing the sense that it is a well-established theory. All experienced practitioners of general theoretical syntax probably know (perhaps unconsciously) that this is not the case. But this does not seem to be widely known across the discipline, so I wrote this paper to emphasize the need for testing the theory by the usual comparative methods (as in §3-4), or to be much more cautious in our assumptions of what the innate grammatical building blocks are.

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